

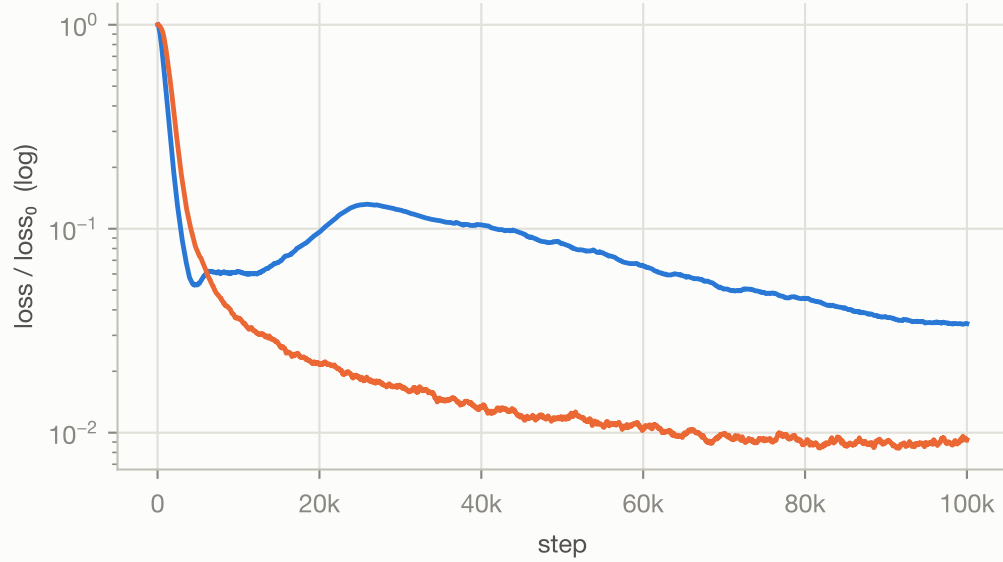
vJEPA vs vMAE — Shear Flow

4ch · 256×512 · incompressible Navier–Stokes shear layer

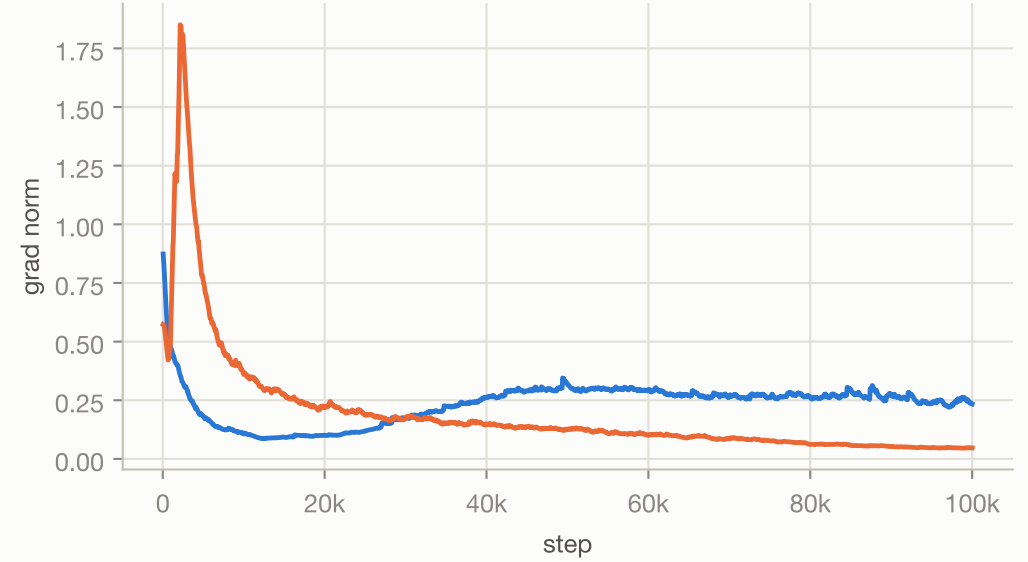
100,000 steps each · identical ViT-S encoder, tube masking @0.9 — wall-clock: JEPA 10.8h / MAE 6.5h — final loss (raw): JEPA 0.016 / MAE 0.012

— JEPA (latent prediction)
— MAE (pixel reconstruction)

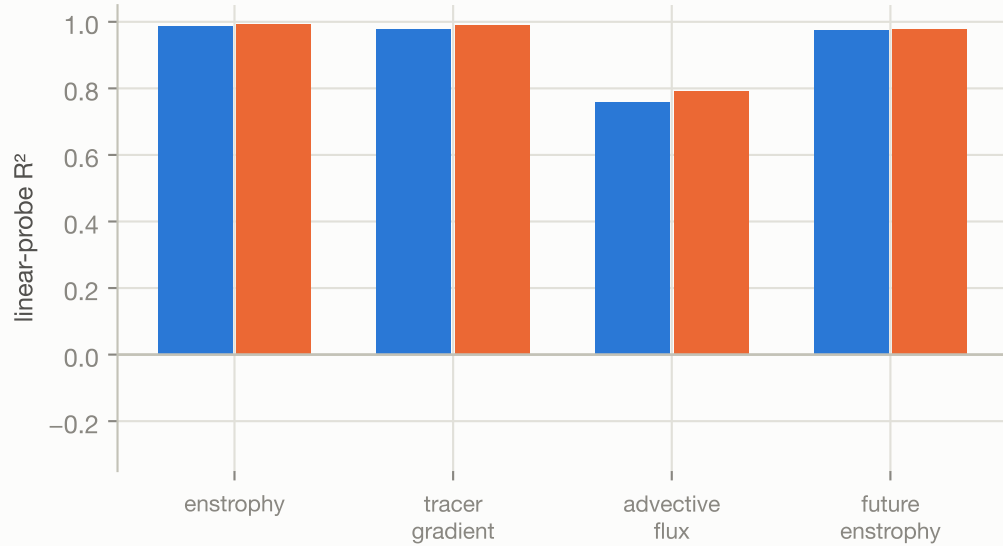
Training loss, relative to step 50



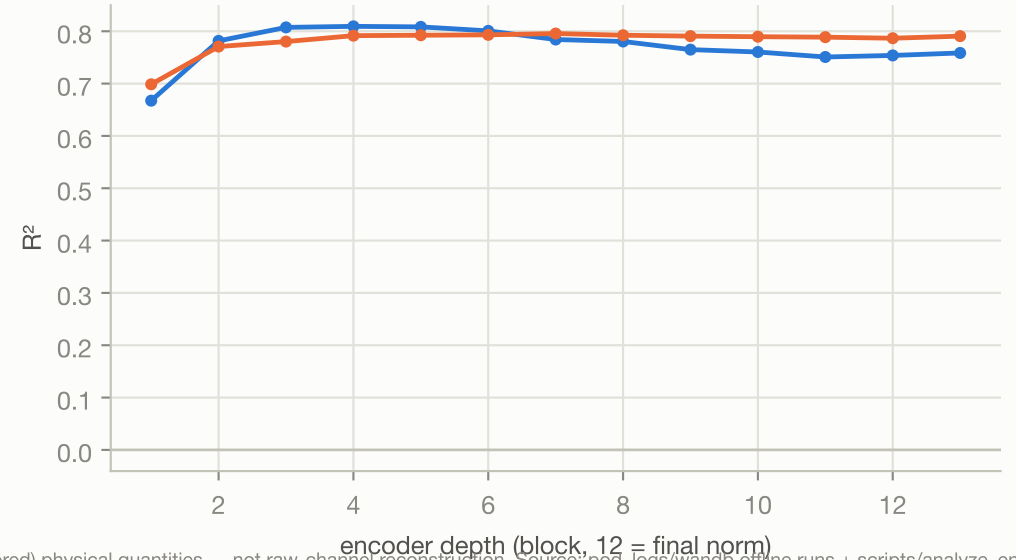
Gradient norm (training stability)



Physics decodability, final layer (ridge probe, 5-fold)



“advective flux” R² by depth



Loss curves EMA-smoothed ($\alpha=0.06$). R² = 5-fold ridge probe on mean-pooled encoder features vs. derived (non-stored) physical quantities — not raw-channel reconstruction. Source: pod_logs/wandb offline runs + scripts/analyze_encoder